

Building Fair and Secure Critical Mineral Supply Chains for the Energy Transition

Prepared for:

The United States Senate Committee on Energy and Natural Resources

Mahasti Yuanita

Master of Public Administration

PUBPOL 5441: Effective Writing for Public Policy

Professor Dan Lamb

March 14, 2026

Executive Summary

The United States Senate Committee on Energy and Natural Resources should authorize a comprehensive strategic mineral partnership program to diversify critical mineral supply chains and reduce dependence on a single foreign supplier for the materials essential to the clean energy transition.

China controls 60 to 80 percent of global refining capacity for lithium, cobalt, nickel, and rare earth elements—the minerals that power electric vehicles, battery storage, and wind turbines. The International Energy Agency (IEA) projects that demand for these minerals will increase nearly fourfold by 2040 under net-zero scenarios, with lithium demand alone rising more than fortyfold (IEA, 2021). Current domestic production cannot meet this demand under any scenario.

This paper recommends three actions:

- Authorize and fund bilateral mineral partnership agreements through an expanded Minerals Security Partnership framework;
- Mandate enforceable supply chain transparency and responsible sourcing standards for federally subsidized mineral purchases; and
- Direct the United States International Development Finance Corporation (DFC) and the Export-Import Bank to prioritize midstream mineral processing investment in partner countries.

These actions will diversify supply chains, reduce geopolitical risk, and support economic development in mineral-producing nations—securing the material foundation for United States clean energy goals.

The Strategic Role of Critical Minerals

Clean energy technologies require substantially more minerals per unit of energy than the fossil fuel infrastructure they replace. The IEA (2021) reports that an electric vehicle requires approximately six times more mineral inputs than a conventional vehicle, and an offshore wind facility requires roughly thirteen times more minerals per unit of electricity than a gas-fired power plant. Lithium, cobalt, nickel, manganese, and rare earth elements are the primary inputs for EV batteries, wind turbines, and electric motors.

The strategic importance of these materials extends well beyond energy. The United States Geological Survey (USGS) classifies 50 minerals as critical to national security and economic competitiveness (USGS, 2024). Executive Order 13953, signed in 2020, declared a national emergency tied to dependence on foreign mineral supply chains (Executive Office of the President, 2020). The Inflation Reduction Act of 2022 reinforced this priority by linking clean energy tax credits to domestic and partner-country mineral sourcing (United States Congress, 2022).

Despite this recognized importance, the United States lacks a coherent, fully funded strategy to develop the international supply relationships needed to meet projected mineral demand. The current patchwork of executive actions, bilateral agreements, and investment tools falls short of what the scale of the challenge demands.

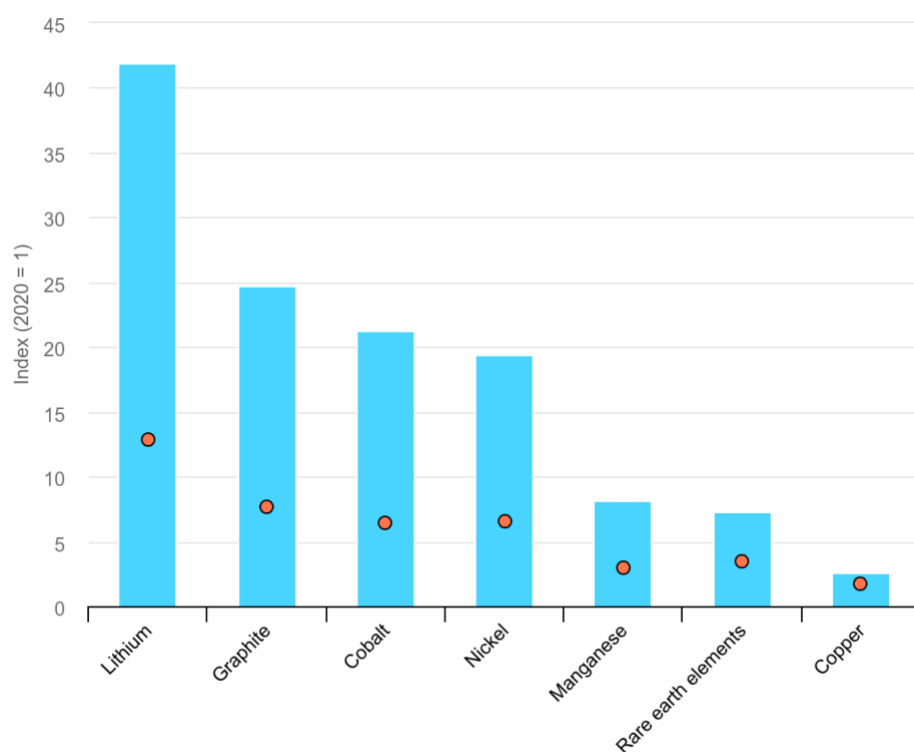
Rising Global Demand

Projected mineral demand illustrates the urgency of action. As Figure 1 shows, demand for lithium could increase more than fortyfold by 2040 relative to 2020 levels under net-zero emissions

scenarios. Demand for cobalt, graphite, and nickel is also expected to rise sharply as electric vehicle adoption accelerates across North America, Europe, and Asia (IEA, 2021).

Figure 1

Projected Growth in Demand for Key Critical Minerals by 2040 Relative to 2020



Note. Adapted from The Role of Critical Minerals in Clean Energy Transitions, by International Energy Agency, 2021 (<https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions>).

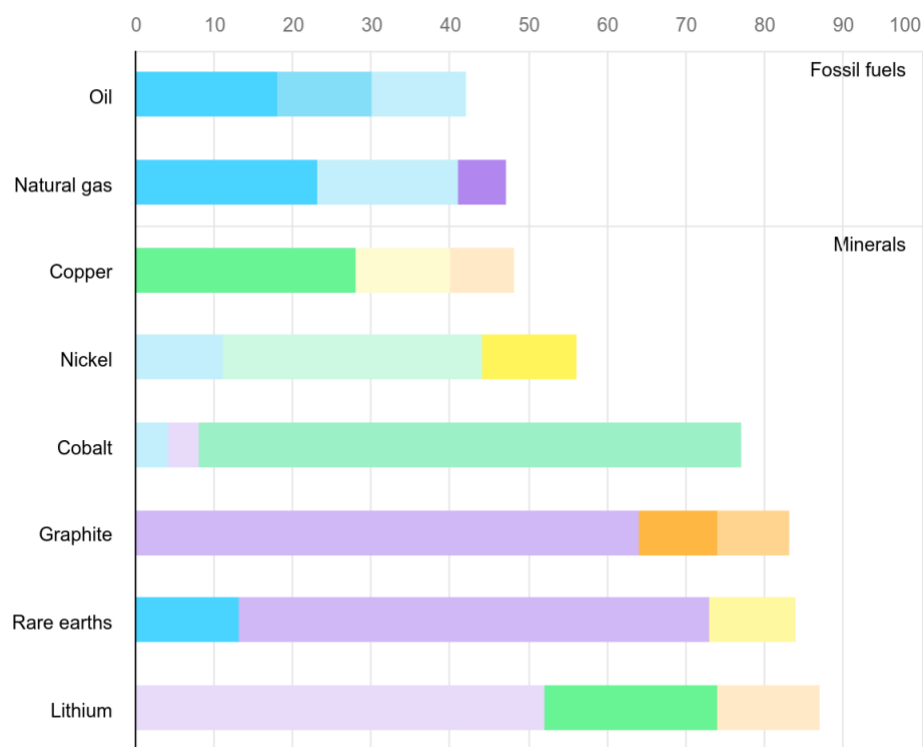
The World Bank (2020) projects that production of minerals including graphite, lithium, and cobalt could increase by nearly 500 percent by 2050 relative to 2018 levels. Current investment in new mining and processing capacity is insufficient to meet this trajectory. Without early action to secure supply partnerships, the clean energy transition faces a structural bottleneck at the very minerals it depends on most.

Supply Chain Concentration and Geopolitical Risk

As Figure 2 shows, production and processing of critical minerals are highly concentrated in a small number of countries. The Democratic Republic of the Congo (DRC) produces approximately 70 percent of global cobalt. Australia, Chile, and Argentina dominate lithium extraction. China controls a dominant share of global refining capacity for virtually all critical mineral categories (IEA, 2021).

Figure 2

Global Production Share of Selected Minerals and Fossil Fuels (2019)



Note. Adapted from The Role of Critical Minerals in Clean Energy Transitions, by International Energy Agency, 2021 (<https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions>).

This concentration creates severe supply vulnerabilities. China demonstrated willingness to weaponize mineral access in 2010 when it restricted rare earth exports to Japan during a diplomatic

dispute, triggering global price spikes and supply shortages (Mancheri et al., 2019). In 2023, China imposed export controls on gallium and germanium—key semiconductor and defense inputs—underscoring that mineral access is an active instrument of geopolitical leverage (Congressional Research Service [CRS], 2023).

Supply chain concentration also raises equity concerns. Most economic value in mineral supply chains concentrates in refining, chemical processing, and battery manufacturing—not extraction. Without access to midstream activities, mineral-producing countries, many of which are low- and middle-income economies, capture only a fraction of the value their resources generate. The DRC earned approximately \$1.5 billion from cobalt exports in 2021, while global cobalt-containing battery sales exceeded \$40 billion (World Bank, 2020). This imbalance is both an economic security problem and a matter of global equity.

Policy Recommendations

The Senate Committee on Energy and Natural Resources should authorize three priority policy actions to diversify mineral supply chains and reduce the United States' strategic vulnerability.

1. Authorize and Expand the Minerals Security Partnership Framework

The Minerals Security Partnership (MSP), launched in 2022 with 14 partner countries, provides the right foundation but lacks sufficient funding and binding commitments (United States Department of State, 2022). Congress should authorize dedicated MSP appropriations and direct the State Department to negotiate bilateral mineral partnership agreements that include joint investment in extraction, refining, and workforce development.

These agreements should incorporate enforceable standards for environmental management, community consultation, and anti-corruption measures. Coordination with the Quad, G7, and other multilateral frameworks will multiply impact and prevent supply gaps from being filled by less transparent actors. Strong partnerships reduce the cost of capital for responsible projects and provide the regulatory predictability that long-term investors require.

2. Mandate Supply Chain Transparency and Responsible Sourcing Standards

Congress should establish enforceable supply chain transparency requirements as a condition of clean energy tax credits under the Inflation Reduction Act. Companies claiming tax benefits for electric vehicle batteries and renewable energy equipment should demonstrate that their mineral inputs meet independently verified standards for environmental management, labor rights, and governance.

Clear, internationally harmonized standards address a documented accountability gap. Amnesty International (2023) documented persistent child labor and unsafe conditions in artisanal cobalt mines in the DRC. Traceability systems with independent auditing create market incentives for responsible sourcing and improve conditions for affected communities. They also reduce regulatory and reputational risk for companies operating across the supply chain.

3. Direct Development Finance Institutions to Prioritize Midstream Processing

Congress should direct the DFC and the Export-Import Bank to establish dedicated critical mineral investment windows with streamlined approval processes, targeting refining and processing capacity in partner countries. Expanding processing capacity outside existing hubs improves supply security while enabling producing countries to capture greater economic value.

Public financing should crowd in private capital by reducing political and project risk. Collaboration with the World Bank, regional development banks, and allied-nation development finance institutions multiplies the impact of United States investment. The DFC currently authorizes up to \$60 billion in financing exposure; Congress should clarify that critical mineral processing explicitly qualifies as a national security priority investment (United States Department of the Treasury, 2022).

Table 1

Key Policy Actions to Strengthen Critical Mineral Supply Chains

Policy Area	Policy Action	Expected Outcome
International Partnerships	Authorize and expand Minerals Security Partnership; negotiate bilateral agreements	Diversified supply chains; reduced geopolitical risk
Supply Chain Transparency	Mandate traceability and responsible sourcing standards tied to IRA tax credits	Improved environmental and labor accountability
Investment and Financing	Direct DFC and Ex-Im Bank to fund refining and processing capacity in partner countries	Greater value capture in partner countries; stronger supply security

Note. Author's analysis based on IEA, CRS, and World Bank reports.

Counter-Arguments and Rebuttals

Critics of a partnership-centered strategy raise three serious objections. Each deserves a direct response.

Counter-Argument 1: Focus on Domestic Production

Some analysts argue that the United States should prioritize domestic mineral production over international partnerships. Proponents point to significant untapped domestic lithium reserves in

Nevada and the Carolinas, cobalt deposits in Minnesota and Idaho, and rare earth deposits in California (USGS, 2024). The argument holds that domestic production eliminates vulnerabilities caused by foreign political instability or trade disputes.

Domestic production expansion is necessary and should proceed in parallel—but it cannot substitute for partnerships. Many critical minerals do not exist in domestic quantities sufficient to meet projected demand under any clean energy scenario. Rare earth processing facilities require seven to ten years to permit and build. A domestic-only strategy forfeits the diversification benefits that partnerships provide. The strongest supply chain combines expanded domestic production where viable with strategic international relationships that spread risk across multiple suppliers and geographies.

Counter-Argument 2: United States Financing Cannot Compete with China

A second objection holds that United States-led partnerships cannot match China's state-backed investment capacity. China has offered infrastructure financing, offtake agreements, and processing investments to mineral-rich developing countries at scale—often with fewer conditionalities than Western institutions require (Overland, 2019). Skeptics argue that adding labor, environmental, and governance conditions makes United States-backed projects less attractive to host governments seeking maximum financial return.

This objection underestimates the demand among resource-rich governments for partnerships that deliver genuine economic development rather than export-only arrangements. Zambia, Chile, and Indonesia have actively sought investment partners that support domestic processing and value-added manufacturing (World Bank, 2020). A growing number of governments recognize that China's infrastructure-for-resources model has left many countries with debt obligations and limited economic diversification. Environmental and governance standards also reduce long-term

political and regulatory risk for investors and host governments alike. United States partnerships that prioritize technology transfer and local value creation are genuinely competitive.

Counter-Argument 3: Market Forces Will Self-Correct

A third perspective holds that private sector investment, guided by price signals, will organically diversify mineral supply chains as demand rises. Government-directed industrial policy, in this view, distorts markets and misallocates capital.

Critical mineral supply chains exhibit characteristics that justify targeted policy intervention. High upfront capital requirements, long development lead times, geopolitical risk premiums, and significant information asymmetries all constitute market failures that private investors systematically underprice. The result is chronic under-investment in supply diversification even as strategic risks accumulate. The rare earth supply crisis of 2010 to 2012—when price spikes caused significant economic disruption before market forces slowly corrected course—illustrates the cost of relying exclusively on market mechanisms in strategic mineral sectors (CRS, 2023). Policy intervention is not market distortion when it addresses genuine market failures with national security consequences.

Implementation

Effective implementation requires sequenced action across three phases.

Phase 1: Partnership Authorization (Year 1 to 2)

Congress should pass authorizing legislation for expanded MSP funding and bilateral mineral partnership agreements. The Committee should hold hearings with allied-nation counterparts, mineral-producing country representatives, and private sector stakeholders to establish framework terms. The State Department, in coordination with the DFC and the Department of Energy, should

identify the ten highest-priority producing countries for initial partnership agreements based on supply criticality, governance, and geopolitical alignment.

Phase 2: Investment and Capacity Building (Years 2 to 5)

The DFC should deploy dedicated critical mineral investment windows with streamlined approval timelines, targeting refining and processing infrastructure in partner countries. The Department of Energy should coordinate technology transfer and workforce training programs. The Committee should establish reporting requirements for DFC and Ex-Im Bank to track investments against supply diversification benchmarks.

Phase 3: Monitoring and Accountability (Ongoing)

Annual supply chain risk assessments, coordinated between the USGS and the Department of Energy, should track progress toward diversification targets and identify emerging vulnerabilities. The Committee should require annual public reporting on partnership performance, investment outcomes, and sourcing standard compliance. These assessments should inform future authorization and appropriations decisions.

Conclusion

Critical minerals underpin the global clean energy transition, but current supply chains remain dangerously concentrated and poorly positioned to meet projected demand. China's dominance in mineral refining is not a market outcome to be managed—it is a strategic vulnerability that requires a deliberate policy response.

Strategic mineral partnerships—grounded in responsible sourcing standards, supported by targeted public financing, and oriented toward building processing capacity in partner countries—offer the

most viable path to supply chains that are secure, diversified, and economically equitable. Domestic production expansion and market-based incentives play important complementary roles, but neither alone achieves the diversification and scale the challenge demands.

The Senate Committee on Energy and Natural Resources should act now: authorize expanded partnership frameworks, direct development finance institutions to prioritize midstream mineral investment, and mandate binding supply chain transparency standards. The energy transition cannot succeed without the minerals that make it possible. Securing those minerals—fairly and at scale—is among the most consequential investments the United States can make in the next decade.

References

- Amnesty International. (2023). *Time to recharge: Corporate action and inaction to tackle abuses in the cobalt supply chain*. Amnesty International. <https://www.amnesty.org/en/documents/afr62/7056/2023/en/>
- Congressional Research Service. (2023). *Critical minerals: Background, legislation, and issues (R46362)*. Congressional Research Service. <https://crsreports.congress.gov/product/pdf/R/R46362>
- Executive Office of the President. (2020). *Executive Order 13953: Addressing the threat to the domestic supply chain from reliance on critical minerals from foreign adversaries*. Federal Register, 85(195), 62539–62541.
- International Energy Agency. (2021). *The role of critical minerals in clean energy transitions*. International Energy Agency. <https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions>
- International Energy Agency. (2024). *Global critical minerals outlook 2024*. International Energy Agency. <https://www.iea.org/reports/global-critical-minerals-outlook-2024>
- Mancheri, N. A., Sprecher, B., Bailey, G., Ge, J., & Tukker, A. (2019). *Effect of Chinese policies on rare earth supply chain resilience*. Resources, Conservation and Recycling, 142, 101–112. <https://doi.org/10.1016/j.resconrec.2018.11.017>
- Overland, I. (2019). *The geopolitics of renewable energy: Debunking four emerging myths*. Energy Research & Social Science, 49, 36–40. <https://doi.org/10.1016/j.erss.2018.10.018>
- United States Congress. (2022). *Inflation Reduction Act of 2022 (Pub. L. 117-169)*. <https://www.congress.gov/bill/117th-congress/house-bill/5376>

United States Department of Energy. (2023). *2023 critical materials assessment*. United States Department of Energy. <https://www.energy.gov/cmei/ammto/articles/2023-doe-critical-materials-assessment>

United States Department of State. (2022). *Minerals Security Partnership*. United States Department of State. <https://www.state.gov/minerals-security-partnership/>

United States Department of the Treasury. (2022). *DFC and allied development finance institutions joint statement on critical minerals*. United States Department of the Treasury. <https://home.treasury.gov/news/press-releases/jy0983>

United States Geological Survey. (2024). *Mineral commodity summaries 2024*. United States Geological Survey. <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024.pdf>

World Bank Group. (2020). *Minerals for climate action: The mineral intensity of the clean energy transition*. World Bank. <https://pubdocs.worldbank.org/en/961711588875536384/Minerals-for-Climate-Action-The-Mineral-Intensity-of-the-Clean-Energy-Transition.pdf>

AI Attestation

Grammarly was used for proofreading and grammar review. AI tools assisted with language clarity review. The policy analysis, argument structure, research, and substantive content of this paper reflect the author's original work.

EXTRA CREDIT: Social Media Post

Platform: X (formerly Twitter) | Draft Post:

⚡ China controls 80% of global cobalt refining. Without action, the #EnergyTransition stalls at the mine. New policy paper calls on @SenateEnergy to authorize strategic #CriticalMinerals partnerships—NOW. Three actions. Real supply chain security. bit.ly/MineralChains2026
#CleanEnergy #EVs #EnergyPolicy #SupplyChain

EXTRA CREDIT: Press Release

FOR IMMEDIATE RELEASE

March 14, 2026

MEDIA CONTACT:

Mahasti Yuanita

myuanita@cornell.edu | (607) 229-9963

Cornell Brooks School of Public Policy

**POLICY PAPER WARNS U.S. ENERGY TRANSITION AT RISK WITHOUT
CRITICAL MINERAL SUPPLY CHAIN REFORM**

*Cornell MPA Paper Calls on Senate Energy Committee to Authorize Strategic Mineral
Partnerships, Direct Development Finance to Processing Capacity*

ITHACA, N.Y. (March 14, 2026) — The United States risks a critical supply chain breakdown for the materials that power electric vehicles and renewable energy systems unless Congress acts immediately to diversify mineral sourcing, according to a policy paper released today by Cornell University's Brooks School of Public Policy.

The paper, "Building Fair and Secure Critical Mineral Supply Chains for the Energy Transition," documents that China controls between 60 and 80 percent of global refining capacity for lithium, cobalt, nickel, and rare earth elements. The International Energy Agency projects that demand for these minerals will increase nearly fourfold by 2040 under net-zero scenarios (IEA, 2021).

The paper is prepared for the United States Senate Committee on Energy and Natural Resources and recommends three priority actions: authorizing bilateral mineral partnership agreements through the Minerals Security Partnership framework; mandating enforceable supply chain

transparency standards; and directing the United States International Development Finance Corporation to invest in mineral processing capacity in partner countries.

"Securing the minerals that make the clean energy transition possible is among the most consequential investments the United States can make in the next decade," said Mahasti Yuanita, author of the paper. "China's dominance in mineral refining is a strategic vulnerability that requires a deliberate policy response—not a market outcome we can wait to self-correct."

The paper also addresses concerns from critics who argue the United States should focus exclusively on domestic production or that private markets will self-correct. The paper argues that a domestic-plus-partnership approach is more resilient than either strategy alone, and that targeted policy intervention is justified given documented market failures in strategic mineral investment. The full paper and supporting analysis are available at: bit.ly/MineralChains2026

Mahasti Yuanita is a Master of Public Administration candidate at Cornell University's Brooks School of Public Policy, specializing in energy and environmental policy. The paper was prepared for PUBPOL 5441: Effective Writing for Public Policy.

###